

GPVN-40.5kV Indoor AC High Voltage Vacuum Circuit Breaker

Summary

GPVN-40.5kV indoor vacuum circuit breaker is a product designed and produced by our company and Xi'an High-voltage Electric Apparatus Institute. The VCB is of rated voltage 40.5kV, three-phase and AC 50/60Hz. It applies to mining firms, power house and substation acting as protection and control electric apparatus. It is also applicable in occasions with frequent operation.

Ambient condition

Altitude: 1000m(Standard); can up to 4500m for special ordering;
Ambient temperature: -25°C ~ +45°C ;
Relative humidity: daily average ≤ 95%, monthly average ≤ 90%;
Earthquake intensity: ≤ 8 degree;
Applicable occasions should free from inflammables, explosives, corrosives and severe vibration.



Model and its meaning

GP VN - 40.5 □ / 1250 - 31.5

Rated short-circuit breaking current (kA)
Rated current (A)
Blank: for insulating sleeve type; E: Embedded type;
M: Permanent Magnet type
Rated voltage (kV)
Indoor vacuum circuit breaker
Company feature code

Structure feature

- The arc-extinguish chamber is on the upper part and the mechanism is on the lower part. This structure is convenient for debug.
- Complex insulating structure using air and organic material; Compact dimension and small weight.
- Vacuum arc-extinguish chamber of Cutler-Hammer Company (USA) and domestic ZMD are both applicable for the VCB. Both two kinds of chambers extinguish arc by vertical magnetic field and featuring with low cut-off and good on-off capability with asymmetry.
- Simple spring operation mechanism is free from maintenance within 10000 times of operations.
- Lead-screw propeller, easy and stable operation and good self-locking capability.

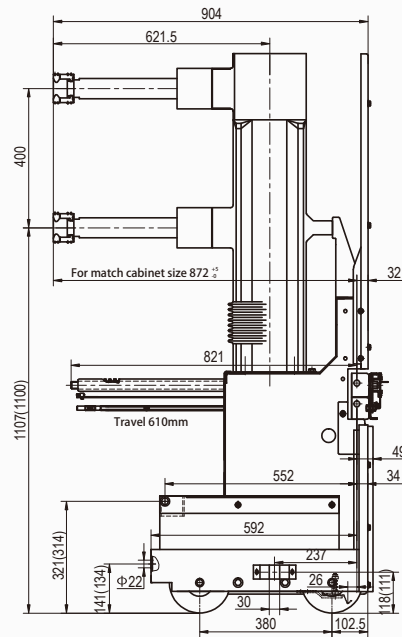
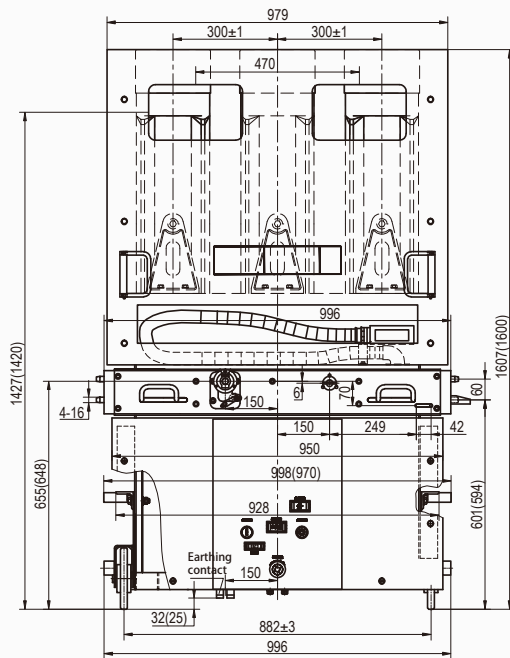
Technical specification

No.	Item	Unit	Data
1	Rated voltage	kV	36/38/40.5
2	Power frequency withstand voltage (1min)	kV	95 (118, isolating distance)
3	Lightning impulse withstand voltage (peak)	kV	185 (215, isolating distance)
4	Rated current	A	630, 1250, 1600, 2000, 2500
5	Rated short-circuit breaking current	kA	20, 25, 31.5
6	Rated short-circuit making current (peak)	kA	50, 63, 80
7	4s Rated short-time withstand current	kA	20, 25, 31.5
8	Rated peak withstand current	kA	50, 63, 80
9	Rated operation sequence		O-0.3s-CO -180s-CO
10	Breaking times of rated short circuit breaking current	times	30
11	Mechanical life	times	10000; 20000 (for magnet type)
12	Rated frequency	Hz	50/60
13	Rated breaking current of capacitor bank	A	400

Technical specification of storage motor of operating mechanism

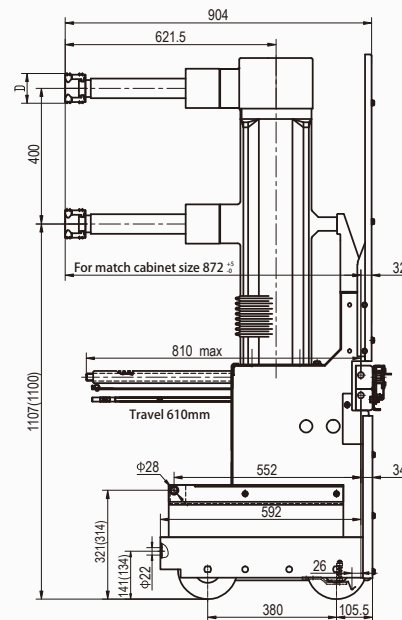
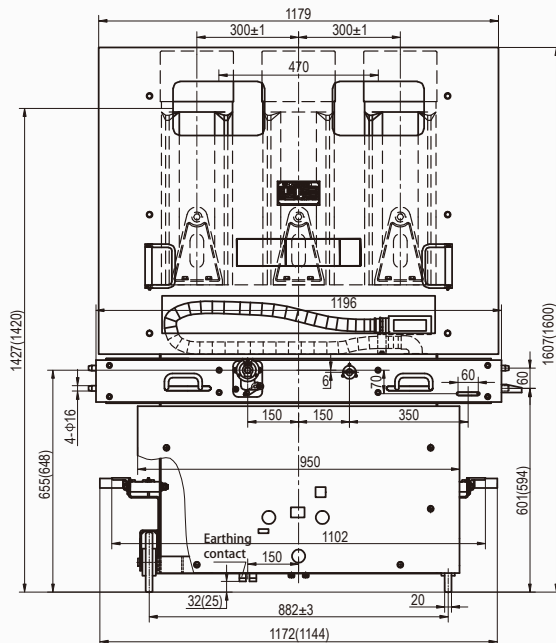
No.	Rated voltage		Rated output power	Normal operate voltage
HDZ-22301B	DC110V AC110V	DC220V AC220V	≤ 230W	85%-110% rated voltage

■ Outline and dimension of installation



Rated current	Matching fixed contact
1250A	Φ 49
1600A	Φ 55
2000A	Φ 79
2500A	Φ 109

GPVN-40.5kV VCB handcart outline dimension for GPN1-40.5kV



Rated current	Matching fixed contact	D
1250A	Φ 49	Φ 89
1600A	Φ 55	Φ 94
2000A	Φ 79	Φ 129
2500A	Φ 109	Φ 159

GPVN-40.5kV VCB handcart outline dimension

Note: 1. Motorized feeding for the VCB is available. Any special additional requirement of application, please contact us for technical disclosure.
2. Here only VCB schematic for reference. For the PT truck, Bus truck, fuse truck, etc. Please consult with GP for more details.

